Instrumenting Node.js code in AWS Lambda

Lambda integrates with AWS X-Ray to help you trace, debug, and optimize Lambda applications. You can use X-Ray to trace a request as it traverses resources in your application, which may include Lambda functions and other AWS services.

To send tracing data to X-Ray, you can use one of two SDK libraries:

- AWS Distro for OpenTelemetry (ADOT) – A secure, production-ready, AWS-supported distribution of the OpenTelemetry (OTel) SDK.
- AWS X-Ray SDK for Node.js – An SDK for generating and sending trace data to X-Ray.

Each of the SDKs offer ways to send your telemetry data to the X-Ray service. You can then use X-Ray to view, filter, and gain insights into your application's performance metrics to identify issues and opportunities for optimization.

> **Important**
>
> The X-Ray and Powertools for AWS Lambda SDKs are part of a tightly integrated instrumentation solution offered by AWS. The ADOT Lambda Layers are part of an industry-wide standard for tracing instrumentation that collect more data in general, but may not be suited for all use cases. You can implement end-to-end tracing in X-Ray using either solution. To learn more about choosing between them, see Choosing between the AWS Distro for Open Telemetry and X-Ray SDKs.

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Using ADOT to instrument your Node.js functions

ADOT provides fully managed Lambda layers that package everything you need to collect telemetry data using the OTel SDK. By consuming this layer, you can instrument your Lambda functions without having to modify any function code. You can also configure your layer to do custom initialization of OTel. For more information, see Custom configuration for the ADOT Collector on Lambda in the ADOT documentation.

For Node.js runtimes, you can add the **AWS managed Lambda layer for ADOT Javascript** to automatically instrument your functions. For detailed instructions on how to add this layer, see AWS Distro for OpenTelemetry Lambda Support for JavaScript in the ADOT documentation.

Using the X-Ray SDK to instrument your Node.js functions

To record details about calls that your Lambda function makes to other resources in your application, you can also use the AWS X-Ray SDK for Node.js. To get the SDK, add the `aws-xray-sdk-core` package to your application's dependencies.

Example blank-nodejs/package.json

```
{
  "name": "blank-nodejs",
  "version": "1.0.0",
  "private": true,
  "devDependencies": {
    "jest": "29.7.0"
  },
  "dependencies": {
    "@aws-sdk/client-lambda": "3.345.0",
    "aws-xray-sdk-core": "3.5.3"
  },
  "scripts": {
    "test": "jest"
  }
}
```

To instrument AWS SDK clients in the AWS SDK for JavaScript v3, wrap the client instance with the `captureAWSv3Client` method.

Example blank-nodejs/function/index.js – Tracing an AWS SDK client

```
const AWSXRay = require('aws-xray-sdk-core');
const { LambdaClient, GetAccountSettingsCommand } = require('@aws-sdk/client-lambda');

// Create client outside of handler to reuse
const lambda = AWSXRay.captureAWSv3Client(new LambdaClient());

// Handler
exports.handler = async function(event, context) {
    event.Records.forEach(record => {
  ...
```

The Lambda runtime sets some environment variables to configure the X-Ray SDK. For example, Lambda sets `AWS_XRAY_CONTEXT_MISSING` to `LOG_ERROR` to avoid throwing runtime errors from the X-Ray SDK. To set a custom context missing strategy, override the environment variable in your function configuration to have no value, and then you can set the context missing strategy programmatically.

Example initialization code

```
const AWSXRay = require('aws-xray-sdk-core');

// Configure the context missing strategy to do nothing
AWSXRay.setContextMissingStrategy(() => {});
```

For more information, see the section called “Environment variables”.

After you add the correct dependencies and make the necessary code changes, activate tracing in your function's configuration via the Lambda console or the API.

Activating tracing with the Lambda console

To toggle active tracing on your Lambda function with the console, follow these steps:

To turn on active tracing

1. Open the Functions page of the Lambda console.
2. Choose a function.
3. Choose **Configuration** and then choose **Monitoring and operations tools**.

4. Under **Additional monitoring tools**, choose **Edit**.
5. Under **CloudWatch Application Signals and AWS X-Ray**, choose **Enable** for **Lambda service traces**.
6. Choose **Save**.

Activating tracing with the Lambda API

Configure tracing on your Lambda function with the AWS CLI or AWS SDK, use the following API operations:

- UpdateFunctionConfiguration
- GetFunctionConfiguration
- CreateFunction

The following example AWS CLI command enables active tracing on a function named **my-function**.

```
aws lambda update-function-configuration --function-name my-function \
--tracing-config Mode=Active
```

Tracing mode is part of the version-specific configuration when you publish a version of your function. You can't change the tracing mode on a published version.

Activating tracing with CloudFormation

To activate tracing on an `AWS::Lambda::Function` resource in an CloudFormation template, use the `TracingConfig` property.

Example function-inline.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Lambda::Function
    Properties:
      TracingConfig:
        Mode: Active
      ...
```

For an AWS Serverless Application Model (AWS SAM) `AWS::Serverless::Function` resource, use the `Tracing` property.

Example template.yml – Tracing configuration

```
Resources:
  function:
    Type: AWS::Serverless::Function
    Properties:
      Tracing: Active
      ...
```

Interpreting an X-Ray trace

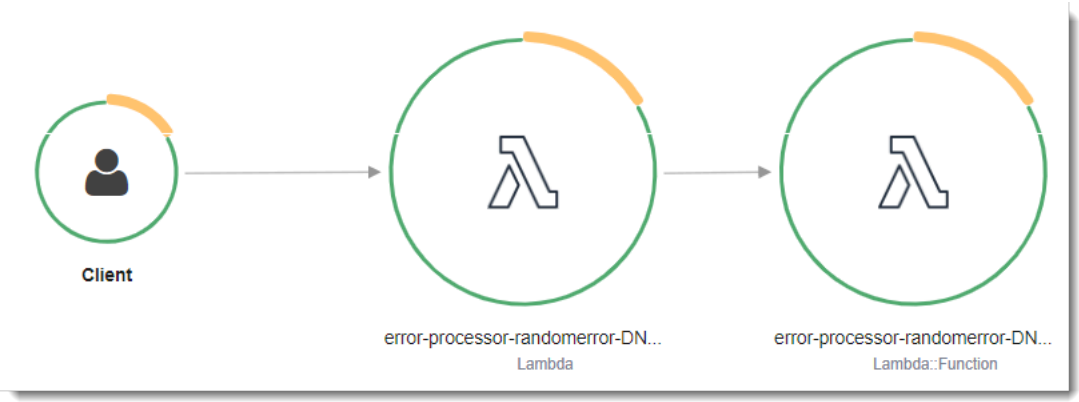
Your function needs permission to upload trace data to X-Ray. When you activate tracing in the Lambda console, Lambda adds the required permissions to your function's execution role. Otherwise, add the AWSXRayDaemonWriteAccess policy to the execution role.

After you've configured active tracing, you can observe specific requests through your application. The X-Ray service graph shows information about your application and all its components. The following example shows an application with two functions. The primary function processes events and sometimes returns errors. The second function at the top processes errors that appear in the first's log group and uses the AWS SDK to call X-Ray, Amazon Simple Storage Service (Amazon S3), and Amazon CloudWatch Logs.

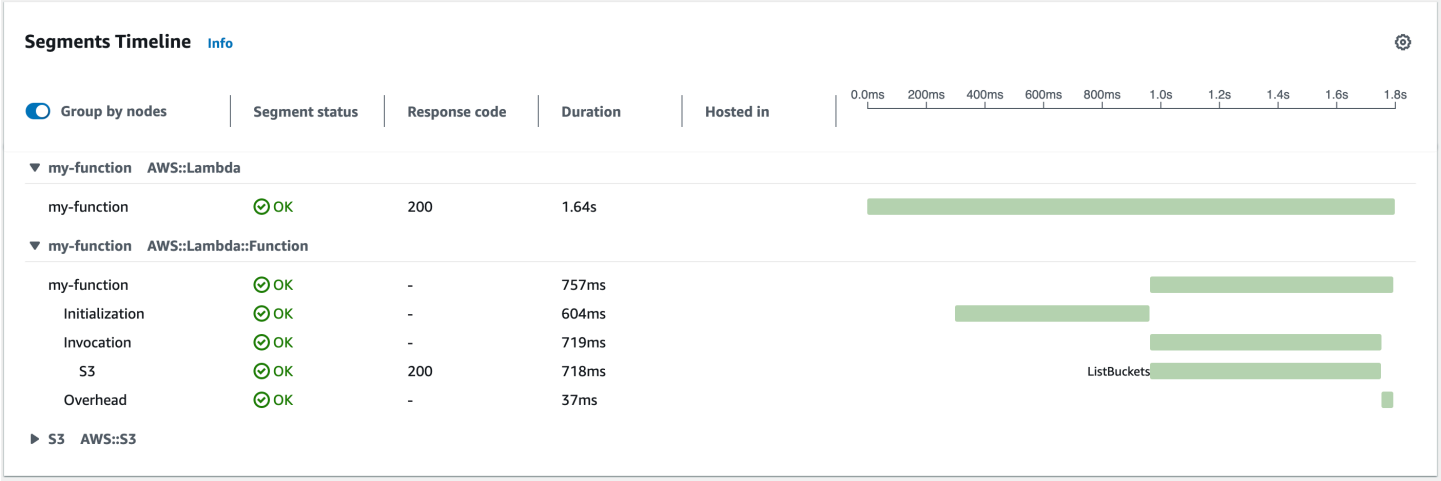
X-Ray doesn't trace all requests to your application. X-Ray applies a sampling algorithm to ensure that tracing is efficient, while still providing a representative sample of all requests. The sampling

rate is 1 request per second and 5 percent of additional requests. You can't configure the X-Ray sampling rate for your functions.

In X-Ray, a *trace* records information about a request that is processed by one or more *services*. Lambda records 2 segments per trace, which creates two nodes on the service graph. The following image highlights these two nodes:



The first node on the left represents the Lambda service, which receives the invocation request. The second node represents your specific Lambda function. The following example shows a trace with these two segments. Both are named **my-function**, but one has an origin of `AWS::Lambda` and the other has an origin of `AWS::Lambda::Function`. If the `AWS::Lambda` segment shows an error, the Lambda service had an issue. If the `AWS::Lambda::Function` segment shows an error, your function had an issue.



This example expands the `AWS::Lambda::Function` segment to show its three subsegments.

> **Note**
>
> AWS is currently implementing changes to the Lambda service. Due to these changes, you may see minor differences between the structure and content of system log messages and trace segments emitted by different Lambda functions in your AWS account.
> The example trace shown here illustrates the old-style function segment. The differences between the old- and new-style segments are described in the following paragraphs.
> These changes will be implemented during the coming weeks, and all functions in all AWS Regions except the China and GovCloud regions will transition to use the new-format log messages and trace segments.

The old-style function segment contains the following subsegments:

- **Initialization** – Represents time spent loading your function and running initialization code. This subsegment only appears for the first event that each instance of your function processes.
- **Invocation** – Represents the time spent running your handler code.
- **Overhead** – Represents the time the Lambda runtime spends preparing to handle the next event.

The new-style function segment doesn't contain an `Invocation` subsegment. Instead, customer subsegments are attached directly to the function segment. For more information about the structure of the old- and new-style function segments, see the section called "Understanding X-Ray traces".

You can also instrument HTTP clients, record SQL queries, and create custom subsegments with annotations and metadata. For more information, see the AWS X-Ray SDK for Node.js in the *AWS X-Ray Developer Guide*.

> **Pricing**
>
> You can use X-Ray tracing for free each month up to a certain limit as part of the AWS Free Tier. Beyond that threshold, X-Ray charges for trace storage and retrieval. For more information, see AWS X-Ray pricing.

Storing runtime dependencies in a layer (X-Ray SDK)

If you use the X-Ray SDK to instrument AWS SDK clients your function code, your deployment package can become quite large. To avoid uploading runtime dependencies every time you update your function code, package the X-Ray SDK in a Lambda layer.

The following example shows an `AWS::Serverless::LayerVersion` resource that stores the AWS X-Ray SDK for Node.js.

Example template.yml – Dependencies layer

```
Resources:
  function:
    Type: AWS::Serverless::Function
    Properties:
      CodeUri: function/.
      Tracing: Active
      Layers:
        - !Ref libs
      ...
  libs:
    Type: AWS::Serverless::LayerVersion
    Properties:
      LayerName: blank-nodejs-lib
      Description: Dependencies for the blank sample app.
      ContentUri: lib/.
      CompatibleRuntimes:
        - nodejs24.x
```

With this configuration, you update the library layer only if you change your runtime dependencies. Since the function deployment package contains only your code, this can help reduce upload times.

Creating a layer for dependencies requires build changes to generate the layer archive prior to deployment. For a working example, see the blank-nodejs sample application.